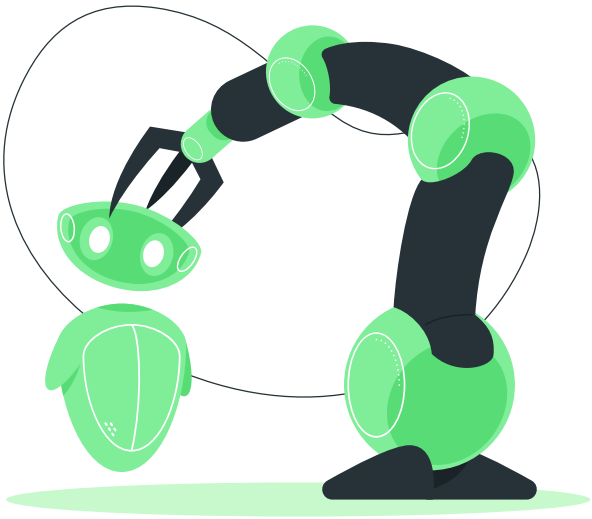
Robotics
CS 3015

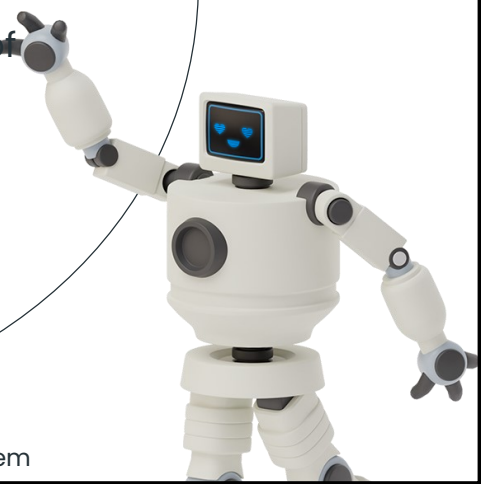
Umarah Qaseem



1

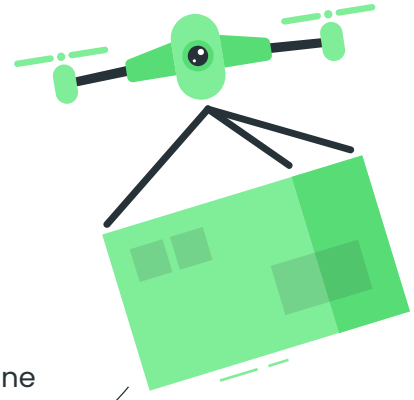
2b

- Pose
- Vector representation
- Vector Transform
- Sequence matters
- Complex Example and Activity of Poses
- Pose Graph
- Derivation of Rotation
- Properties of R
- Derivation of Translation
- Transformation Matrix
- Making everything real



2

What did we learn?



Position of things in the world

Frame of Reference

2D Geometry, Story of Cartesian Plane

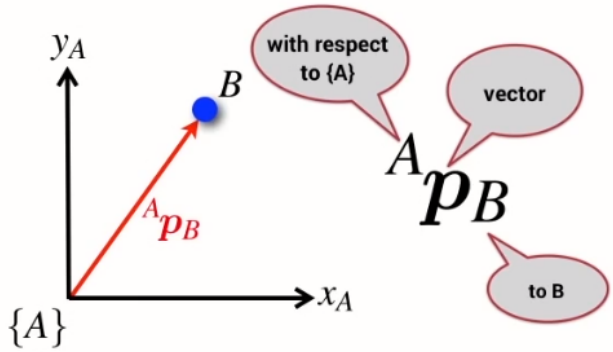
Convention of coordinate frame

Revision

4

Point and Vector

- Point is described by a vector with respect to a coordinate frame.



Revision

5



Pose

Pose = Position and Orientation

Technically: Translation + Rotation

Pronounced as ksi

It has 3 parameters

(x, y, θ)

ξ

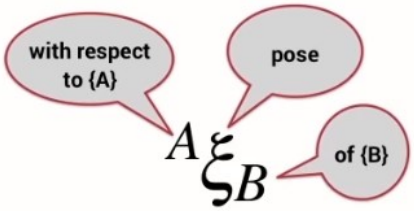
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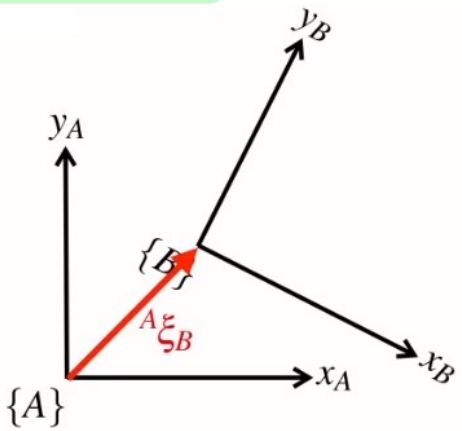
Revision

6

Pose Example

- The motion of a coordinate frame



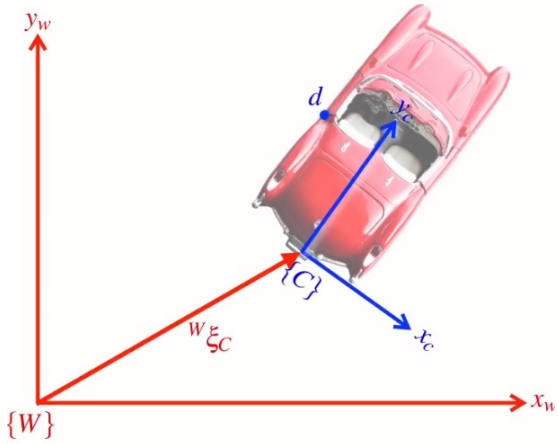


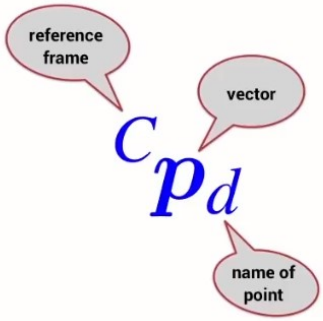
Revision

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7

Vector to point d from Frame C

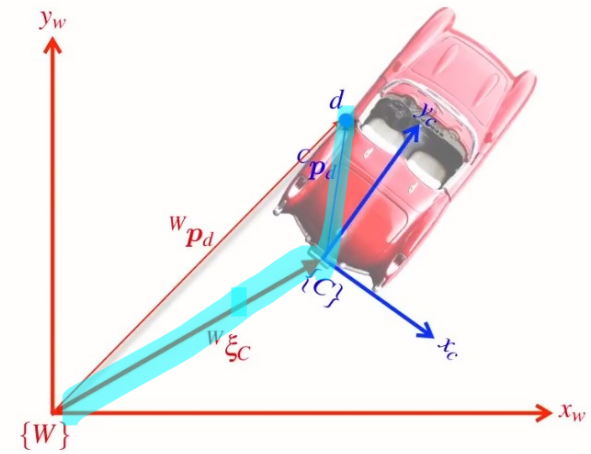




Revision

8

Vector to point d from World Frame



Revision

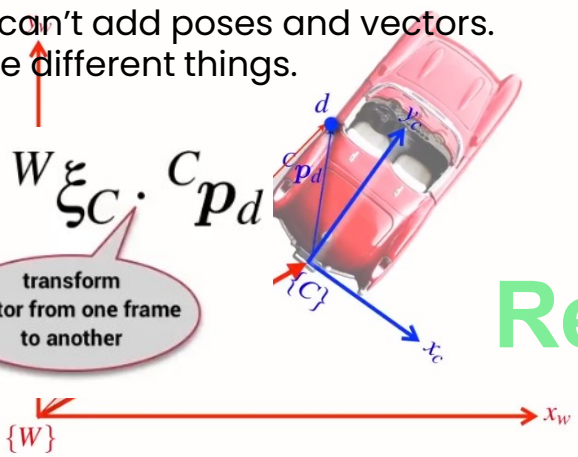
9

Transforming vector from one Frame to another

- But we can't add poses and vectors.
- They are different things.

$$^{W}\boldsymbol{p}_d = {}^{W}\xi_C \cdot {}^{C}\boldsymbol{p}_d$$

transform vector from one frame to another



Revision

10

Relative Pose

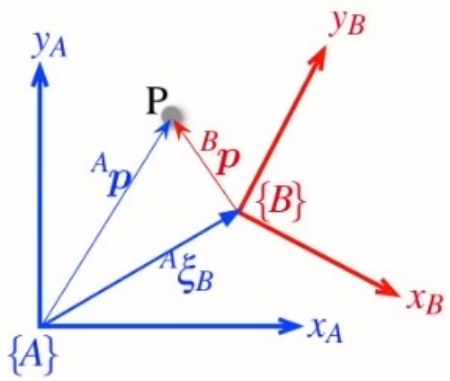
Describing relationships between the frames.

Describing a point, with a vector with respect to any frame.

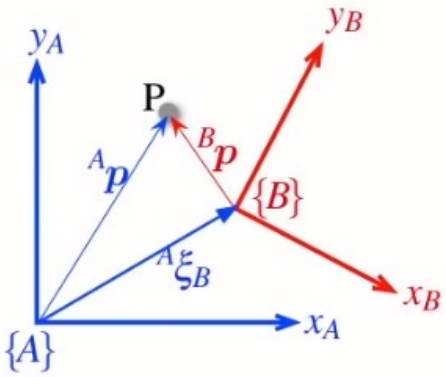


11

Relative points



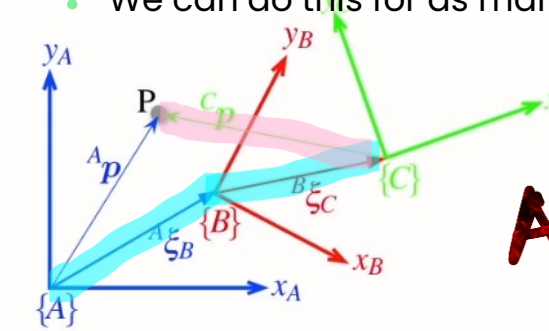
Relative points



$$ {}^A\boldsymbol{p} = {}^A\xi_B \cdot {}^B\boldsymbol{p} $$

Relative points

- Relative pose can be compounded or composed.
- We can do this for as many poses as we want.



composition

$$^A\xi_C = {}^A\xi_B \oplus {}^B\xi_C$$

$$^A p = {}^A\xi_C \cdot {}^C p$$

$$^A\boldsymbol{p} = ({}^A\xi_B \oplus {}^B\xi_C) \cdot {}^C\boldsymbol{p}$$



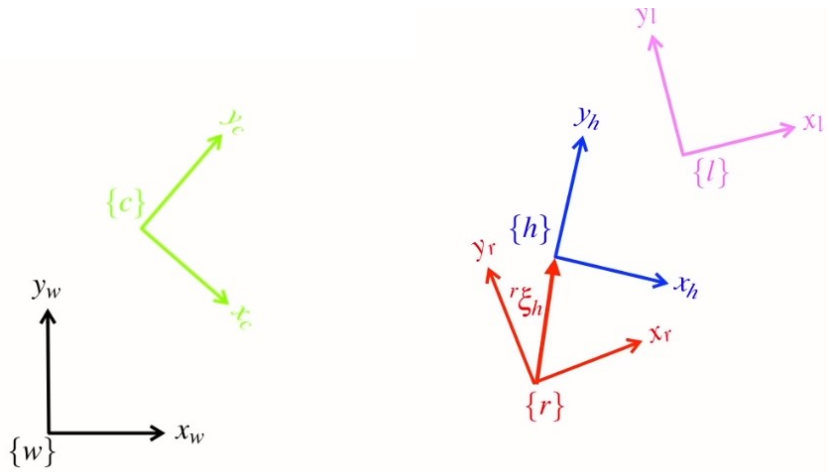
Order Matters!

Illustration

- Ordering matters!
- Tr + rot
- Rot + tr

16

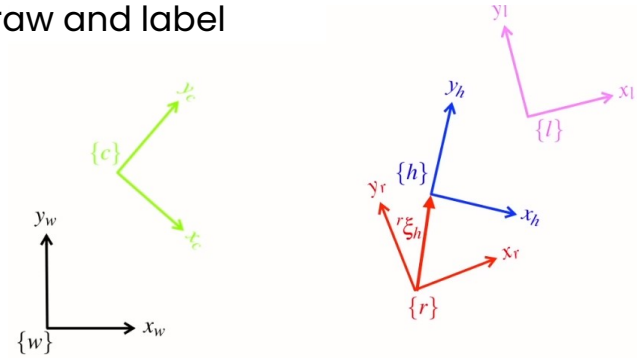
Complex Example



17

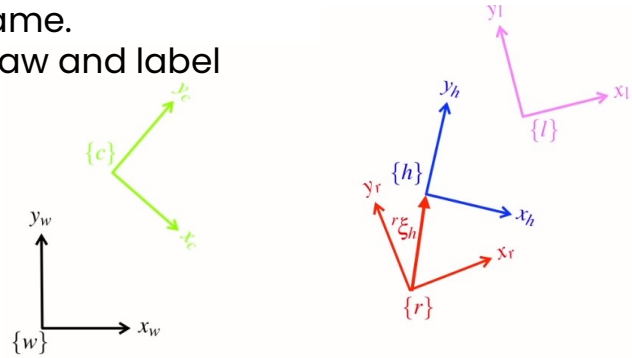
Activity

- Where is robot wrt the camera?
- It means pose of robot frame wrt to camera frame.
- Draw and label

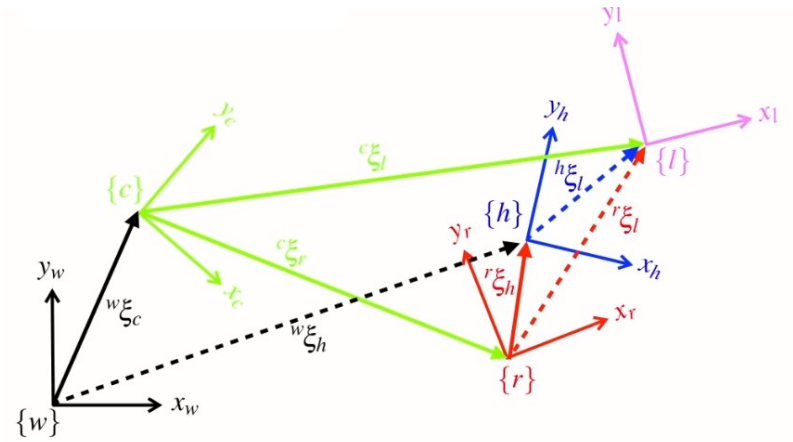


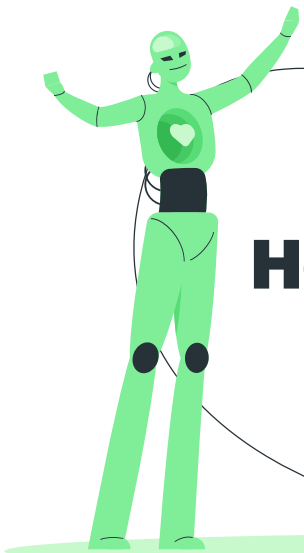
Activity

- Where is little robot wrt the camera?
- It means pose of little robot frame wrt to camera frame.
- Draw and label



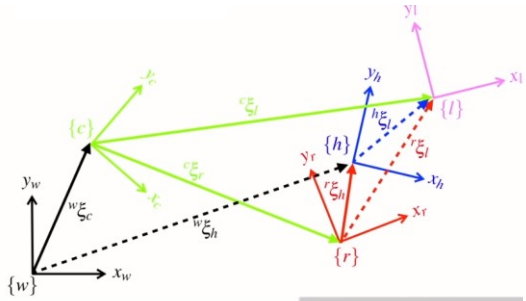
Complex Example





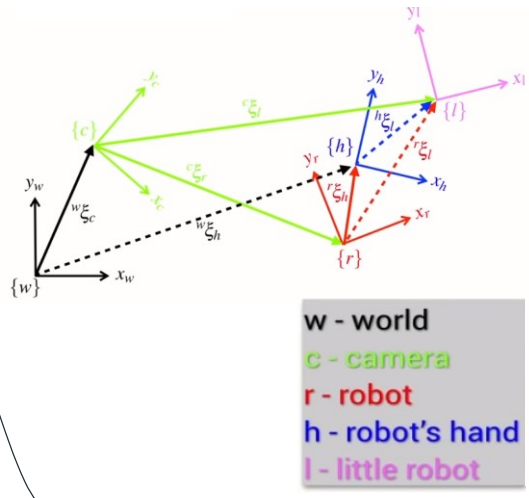
How to make it simple?

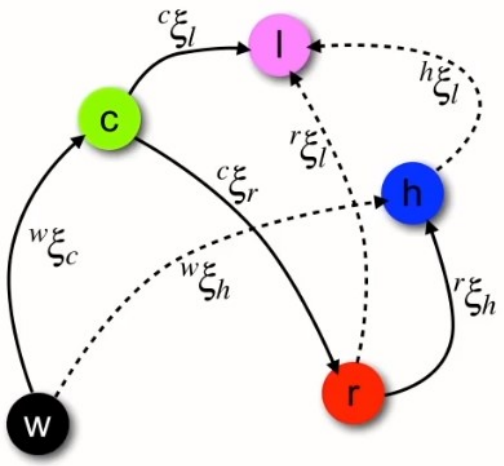
Solution?



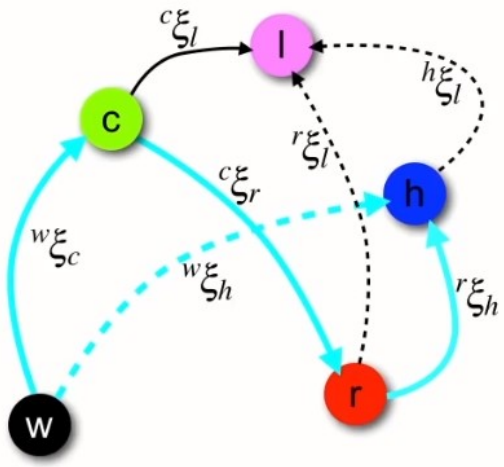
Make a directed pose graph!

Solution – A directed graph!



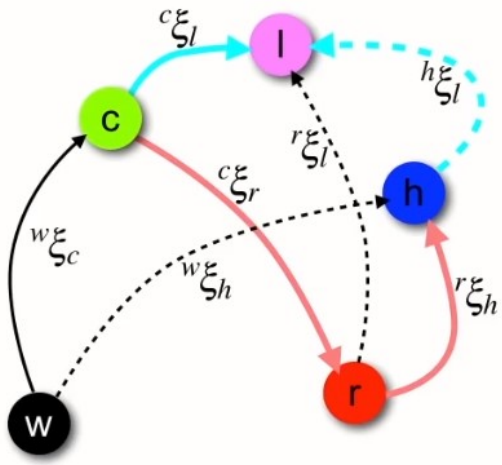


Example 1: Finding required pose in Pose Graph



$$^{w}\xi_{h} = {}^{w}\xi_{c} \oplus {}^{c}\xi_{r} \oplus {}^{r}\xi_{h}$$

Example 2: Finding a required pose – Complex

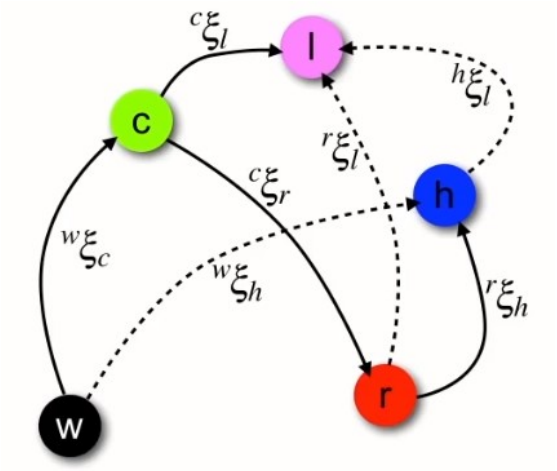


$$^{h}\xi_{l} = \ominus\, {}^{r}\xi_{h} \ominus {}^{c}\xi_{r} \oplus {}^{c}\xi_{l}$$

reverse direction

Example 3: Activity

- Find the third unknown pose yourself!



Pose Algebra

$$X\xi_Y \oplus Y\xi_Z = X\xi_Z$$

$$\xi_1 \oplus \xi_2 \neq \xi_2 \oplus \xi_1$$

$$\ominus {}^X\xi_Y = {}^Y\xi_X$$

$$\xi \ominus \xi = 0 \qquad \ominus\xi \oplus \xi = 0$$

$$\xi \ominus 0 = \xi \qquad \xi \oplus 0 = \xi$$

Pose/Spatial Algebra

$${}^{X}\xi_{Y} \oplus {}^{Y}\xi_{Z} = {}^{X}\xi_{Z}$$

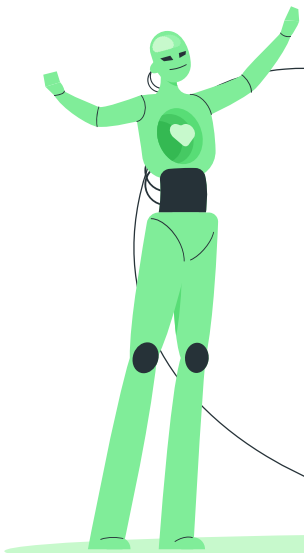
$$\xi_1 \oplus \xi_2 \neq \xi_2 \oplus \xi_1$$

$$\ominus {}^{X}\xi_{Y} = {}^{Y}\xi_{X}$$

$$\xi \ominus \xi = 0 \qquad \ominus\xi \oplus \xi = 0$$

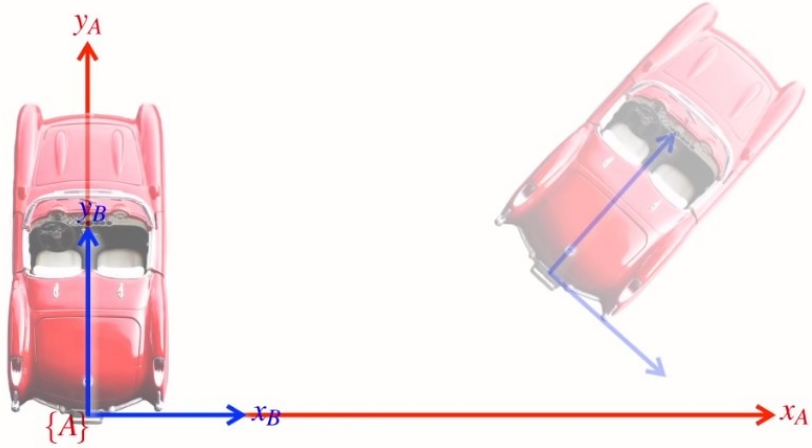
$$\xi \ominus 0 = \xi \qquad \xi \oplus 0 = \xi$$

$${}^{X}p = {}^{X}\xi_{Y} \bullet {}^{Y}p$$



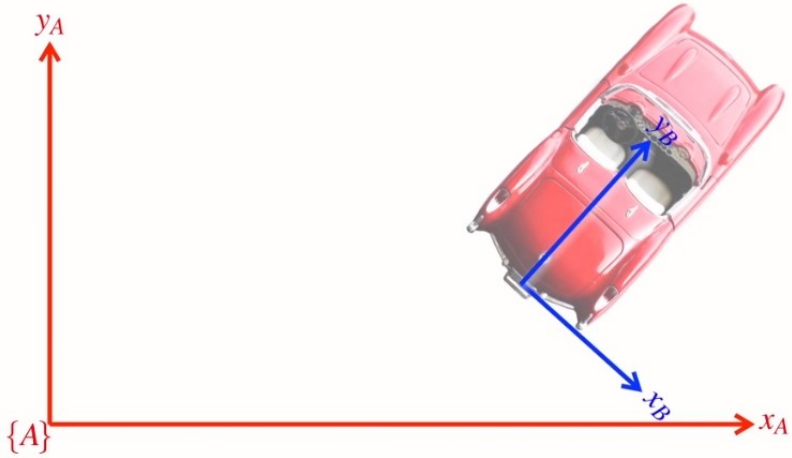
Now let's make it real

Real Representation of pose in 2D



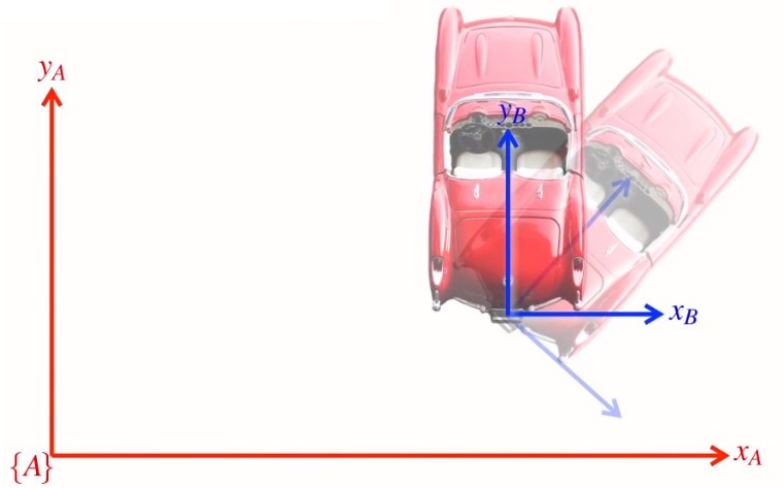
30

Real Representation of pose in 2D



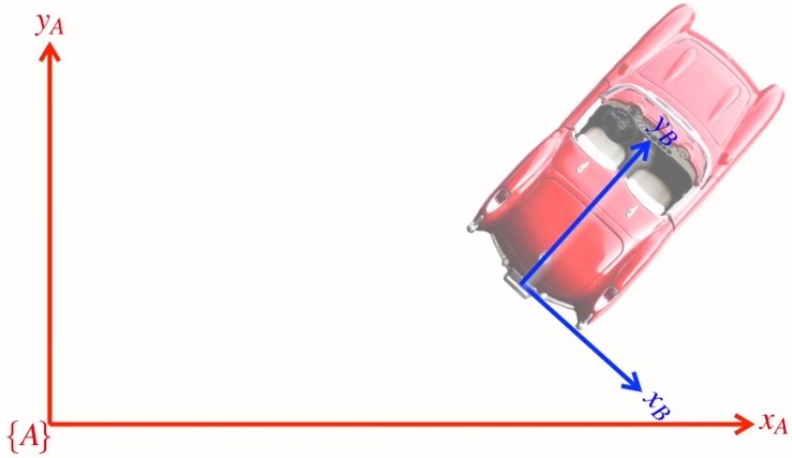
31

1- Translation



32

2- Rotation



33

Transformation

Mathematics and derivation



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34

Derivation

Rotation

35

Derivation

Solved on board

36

Rotation Matrix

$$\mathbf{R} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

37

Rotation Matrix

$$ {}^{U}\mathbf{R}_{T} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} $$

38

Rotation

- R is a function of rotation angle from Frame U to Frame T.
- We can also say, R rotates vectors from Frame T to Frame U

$$ {}^{U}\mathbf{R}_{T} $$

39

Properties of R

- R is an orthogonal (orthonormal) matrix
  - Each column is a unit length vector
  - Each column is orthogonal to all other columns i.e. dot product will be zero.
- The inverse is same as the transpose. $\mathbf{R^{-1} = R^T}$
- The determinant is always +1 $\mathbf{det(R) = 1}$
  - This means? If we multiply this matrix with any vector, then the Length of vector is unchanged by rotation!

40

Inverse Rotation

- $\underline{^U p} = {}^U R_T \, {}^T p$
- $^T p = {}^U R_T^{-1} \, \underline{^U p}$

- To rotate inversely i.e., Rotate vector from frame A to frame B (rotation angle from B to A)
- Inverse Rotation matrix will be used.
- Inverse is equal to transpose

- $^T p = {}^T R_U \, \underline{^U p}$
- $^T R_U = {}^U R_T^{-1}$

41

Derivation

Adding Translation into picture

42

Derivation

Solved on board

43

Transformation

homogeneous vector

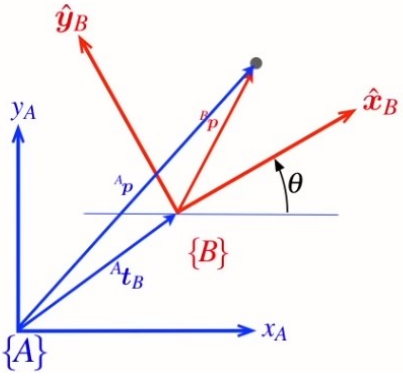
homogeneous transform

homogeneous vector

$$^{A}\tilde{\boldsymbol{p}} = \begin{pmatrix} ^{A}\mathbf{R}_{B} & ^{A}\boldsymbol{t}_{B} \\ 0,0 & 1 \end{pmatrix} {}^{B}\tilde{\boldsymbol{p}}$$

$$^{A}\tilde{\boldsymbol{p}} = {}^{A}\mathbf{T}_{B}\,{}^{B}\tilde{\boldsymbol{p}}$$

$$^{A}\mathbf{T}_{B} = \begin{pmatrix} ^{A}\mathbf{R}_{B} & ^{A}\boldsymbol{t}_{B} \\ 0,0 & 1 \end{pmatrix}$$



$$\boldsymbol{p} = \begin{pmatrix} a \\ b \end{pmatrix} \qquad \tilde{\boldsymbol{p}} = \begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$$

44

Let's make things real

- Pose is just a Matrix!

$$^{A}\xi_{B} \sim {}^{A}\mathbf{T}_{B} = \begin{pmatrix} \cos\theta & -\sin\theta & t_x \\ \sin\theta & \cos\theta & t_y \\ 0 & 0 & 1 \end{pmatrix}$$

- Composition is just a matrix product!

$$^{X}\xi_{Y} \oplus {}^{Y}\xi_{Z} \mapsto {}^{X}\mathbf{T}_{Y}\,{}^{Y}\mathbf{T}_{Z}$$

- Negation is a matrix Inverse!

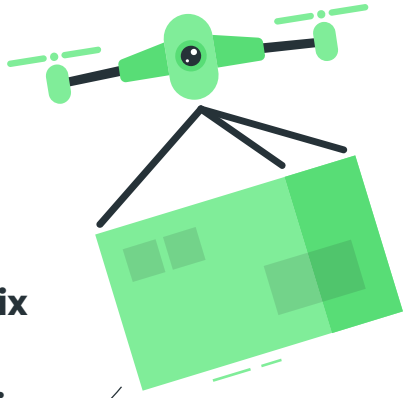
$$\ominus\xi \mapsto \mathbf{T}^{-1}$$

- Inverse of T is NOT its transpose!
- Vector transformation is a matrix-vector product
- Remember dot operator?
- (which transforms vector from one Frame to another)

$$\xi \cdot \boldsymbol{p} \mapsto \mathbf{T}\tilde{\boldsymbol{p}}$$

45

Home Task



Study what is a homogeneous matrix
Orthonormal, orthogonal
Install MATLAB and Robotic tool box.

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48